COMPACT PUSHFORWARD ALONG THE AFFINE LINE OF A TORSION CONSTRUCTIBLE SHEAF HAS CONSTRUCTIBLE COHOMOLOGY

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Constructible sheaves as a colimit (Etale Cohomology, Lemma etale-cohomology-lemma-category-is-colimit, canonical tag 09YU). cited at proof lines 4882-4886. Lets the constructible sheaf F descend along the limit presentation of Y, giving the Noetherian finite dimensional reduction.
- Relative triangle associated to an open (Lemma lemma-relative-triangle-associated-to-open, canonical tag 0GKP). cited at proof lines 4919-4920. The distinguished triangle for U in the affine line that discards the complementary part Z.
- Locally constant case (Lemma lemma-loc-constant-shriek-rel-dim-1, canonical tag 0GKY). cited at proof line 4924. The terminal named case: R(U to Y)_! of the finite type locally constant restriction has constructible cohomology.

Source reference. Original lemma 09YU: sources/stacks-chow/tags.txt, source lines 12577–12577. Cited at proof lines 4884-4886: F is the pullback of a constructible sheaf on A^1 over some finite type Z-subalgebra, giving the Noetherian reduction.

Source reference. Original lemma 0F7L: sources/stacks-chow/tags.txt, source lines 19011–19011. Cited at proof lines 4906-4907: base change compatibility of Rf_!F handles the complement in the induction step.

Source reference. Original lemma 0GKR: sources/stacks-chow/tags.txt, source lines 20696–20696. Cited at proof line 4899: finiteness of H^q_c of the fibre in the base case d = 0.

Source reference. Original lemma 0GKY: sources/stacks-chow/tags.txt, source lines 20703–20703. Cited at proof line 4924: the terminal case, constructibility of R(U to Y)_!F for the locally constant restriction.

Source reference. Original lemma 0GK3: sources/stacks-chow/tags.txt, source lines 20673–20673. Cited at proof lines 4921-4922: the finite complement Z to Y has constructible Rf_!, discarding it from the triangle.

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Source reference. Original lemma 0GKP: sources/stacks-chow/tags.txt, source lines 20694–20694. Cited at proof lines 4919-4920: the distinguished triangle for the open U in the affine line.

Source reference. Original lemma 0GK0: sources/stacks-chow/tags.txt, source lines 20670–20670. Cited at proof lines 4896-4898: identifies the stalks of R^qf_!F with the compact cohomology of the geometric fibre in the base case.

Source reference. Original lemma 02NW: sources/stacks-chow/tags.txt, source lines 3495–3495. Cited at proof lines 4916-4918: after shrinking Y to a dense open, Z to Y is finite.

Lemma 0.2. *Let Y be an affine scheme. Let Λ be a Noetherian ring. Let $\mathcal{F}$ be a constructible sheaf of Λ-modules on $\mathbf{A}^1_Y$ which is torsion. Then $Rf_!\mathcal{F}$ has constructible cohomology sheaves where $f : \mathbf{A}^1_Y \to Y$ is the structure morphism.*

Proof. Say $\mathcal{F}$ is annihilated by $n > 0$. Then we can replace Λ by $\Lambda/n\Lambda$ without changing $Rf_!\mathcal{F}$. Thus we may and do assume Λ is a torsion ring.

Say $Y = \mathrm{Spec}(R)$. Then, if we write $R = \bigcup R_i$ as the union of its finite type $\mathbf{Z}$-subalgebras, we can find an i such that $\mathcal{F}$ is the pullback of a constructible sheaf of Λ-modules on $\mathbf{A}^1_{R_i}$, see Étale Cohomology, Lemma 09YU. Hence we may assume Y is a Noetherian scheme of finite dimension.

Assume Y is a Noetherian scheme of finite dimension $d = \dim(Y)$ and Λ is torsion. We will prove the result by induction on d.

Base case. If $d = 0$, then the only thing to show is that the stalks of $R^q f_!\mathcal{F}$ are finite Λ-modules. If $\overline{y}$ is a geometric point of Y, then we have $(R^q f_!\mathcal{F})_{\overline{y}} = H^q_c(X_{\overline{y}}, \mathcal{F})$ by Lemma 0GK0. This is a finite Λ-module by Lemma 0GKR.

Induction step. It suffices to find a dense open $V \subset Y$ such that $Rf_!\mathcal{F}|_V$ has constructible cohomology sheaves. Namely, the restriction of $Rf_!\mathcal{F}$ to the complement $Y \setminus V$ will have constructible cohomology sheaves by induction and the fact that formation of $Rf_!\mathcal{F}$ commutes with all base change (Lemma 0F7L). By definition of constructible sheaves of Λ-modules, there is a dense open subscheme $U \subset \mathbf{A}^1_Y$ such that $\mathcal{F}|_U$ is a finite type, locally constant sheaf of Λ-modules. Denote $Z \subset \mathbf{A}^1_Y$ the complement (viewed as a reduced closed subscheme). Note that U contains all the generic points of the fibres of $\mathbf{A}^1_Y \to Y$ over the generic points $\xi_1, \ldots, \xi_n$ of the irreducible components of Y. Hence $Z \to Y$ has finite fibres over $\xi_1, \ldots, \xi_n$. After replacing Y by a dense open (which is allowed), we may assume $Z \to Y$ is finite, see Morphisms, Lemma 02NW. By the distinguished triangle of Lemma 0GKP and the result for $Z \to Y$ (Lemma 0GK3) we reduce to showing that $R(U \to Y)_!\mathcal{F}$ has constructible cohomology sheaves. This is Lemma 0GKY. □